

Practice 3.1 Correlations & Linear Regression

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Correlation

A correlation is where a data has something to do with other; where if one data on a database changes, the other changes either in the same direction or in opposite direction

positive correlation

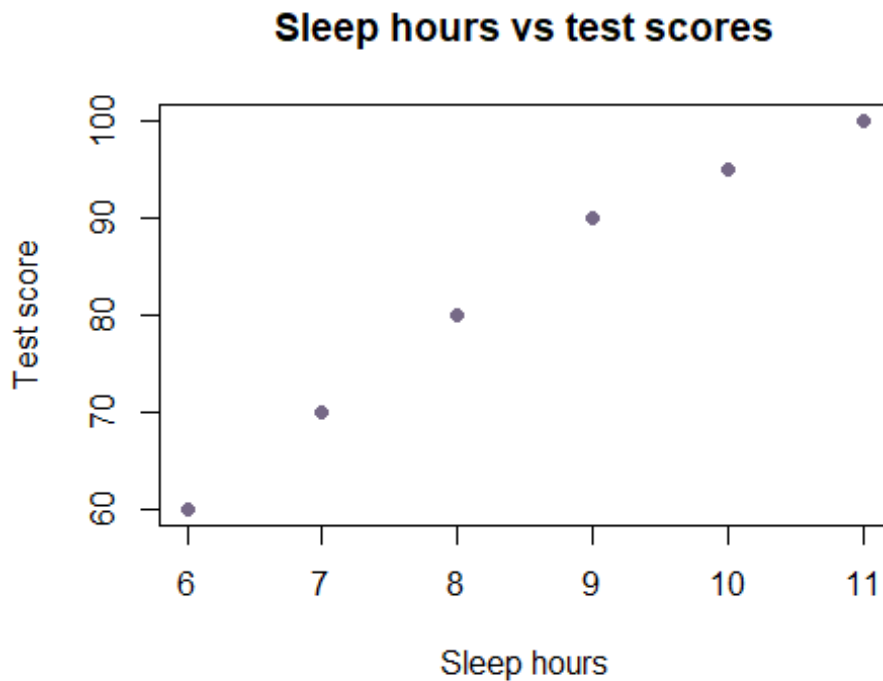
Is where the data varies at the same rate between data

```
sleep_hours <- c(6, 7, 8, 9, 10, 11)
test_score <- c(60, 70, 80, 90, 95, 100)

correlation1 <- cor(sleep_hours, test_score)
cat("The correlation of sleep time and test scores are: ", correlation1)

## The correlation of sleep time and test scores are: 0.9885054

#Dispersion Graphic
plot(sleep_hours, test_score,
     main="Sleep hours vs test scores",
     xlab="Sleep hours",
     ylab="Test score",
     pch=19, col="#766987")
```



Negative correlation

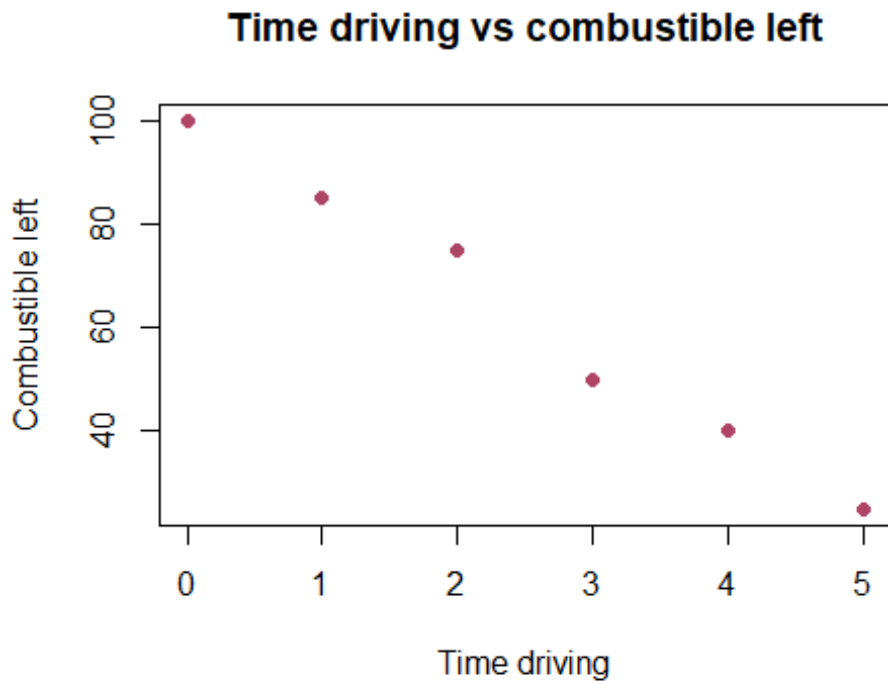
A negative correlation is where the data varies at the opposite rate between data

```
time_driving <- c(0, 1, 2, 3, 4, 5)
combustible_left <- c(100, 85, 75, 50, 40, 25)

correlation2 <- cor(time_driving, combustible_left)
cat("The correlation between the time driving and combustible left is: ",
    correlation2)

## The correlation between the time driving and combustible left is: -
0.994113

#Dispersion graphic
plot(time_driving, combustible_left,
     main="Time driving vs combustible left",
     xlab="Time driving",
     ylab="Combustible left",
     pch=19, col="#af4566")
```



No correlation

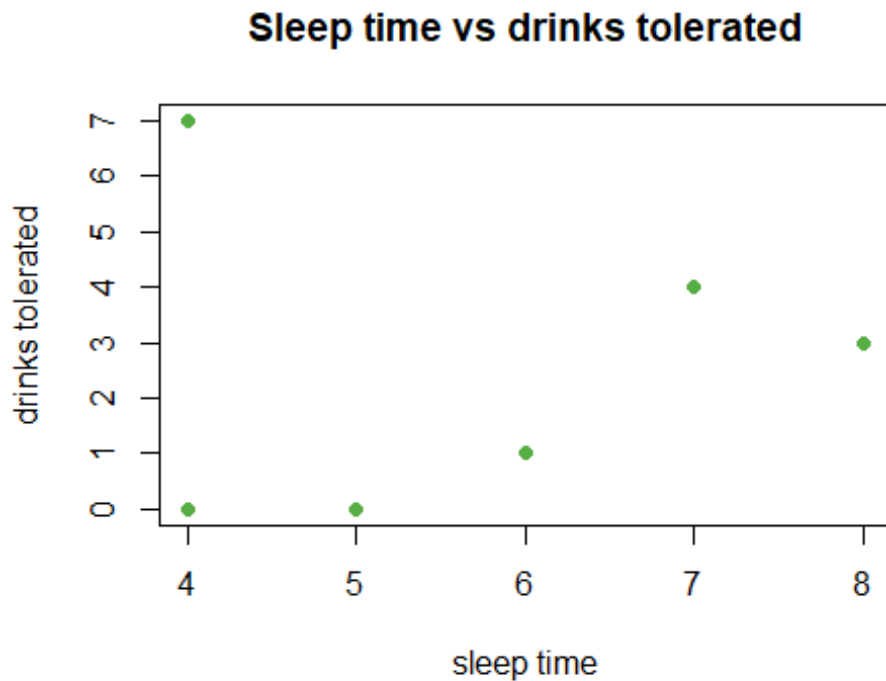
There's no correlation where, while one data modifies, the other either remains the same or varies way too unpredictably

```
rest <- c(6, 4, 7, 8, 4, 5)
drink <- c(1, 0, 4, 3, 7, 0)

correlation3 <- cor(rest, drink)
cat("The correlation between the time sleeping and the drinks tolerated
is:", correlation3)

## The correlation between the time sleeping and the drinks tolerated is:
0.04472136

#Dispersion Graphic
plot(rest, drink,
      main="Sleep time vs drinks tolerated",
      xlab="sleep time",
      ylab="drinks tolerated",
      pch=19, col="#56ad43")
```



MTCARS RELATIONS

#Importing a dataset, correlate the data between miles per gallon and the number of cylinders

```
data("mtcars")
milespergallon <- mtcars$mpg
cylinders <- mtcars$cyl

correlation4 <- cor(milespergallon, cylinders)
cat("The correlation between the miles per gallon and the cylinders of cars is: ", correlation4)

## The correlation between the miles per gallon and the cylinders of cars is: -0.852162

#Dispersion Graphic
plot(milespergallon, cylinders,
     main="Miles per gallon vs number of cylinders",
     xlab="Miles per gallon",
     ylab="Number of cylinders",
     pch=19, col="#3576cd")
```

Miles per gallon vs number of cylinders

